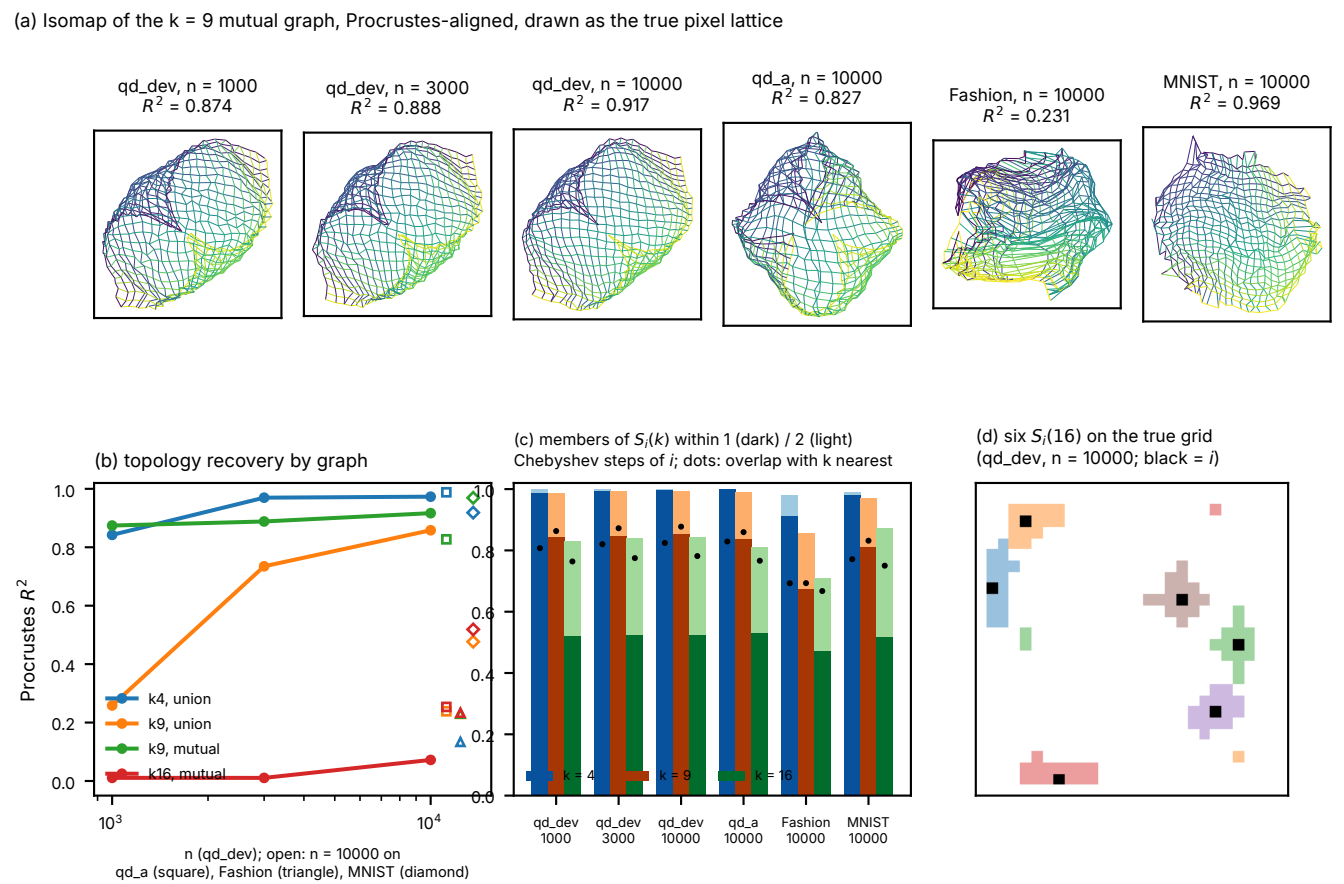

(a) Isomap of the k = 9 mutual graph, Procrustes-aligned, drawn as the true pixel lattice
qd_dev, n = 1000
R^2 = 0.874
qd_dev, n = 3000
R^2 = 0.888
qd_dev, n = 10000
R^2 = 0.917
qd_a, n = 10000
R^2 = 0.827
Fashion, n = 10000
R^2 = 0.231
MNIST, n = 10000
R^2 = 0.969
(b) topology recovery by graph
Procrustes R^2
k4, union
k9, union
k9, mutual
k16, mutual
n (qd_dev); open: n = 10000 on
qd_a (square), Fashion (triangle), MNIST (diamond)
(c) members of $S_i(k)$ within 1 (dark) / 2 (light)
Chebyshev steps of i; dots: overlap with k nearest
k = 4
k = 9
k = 16
qd_dev 1000
qd_dev 3000
qd_dev 10000
qd_a 10000
Fashion 10000
MNIST 10000
(d) six $S_i(16)$ on the true grid
(qd_dev, n = 10000; black = i)